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METHODS AND APPARATUSES FOR DETERMINING TRANSDUCER LOCATIONS TO GENERATE TUMOR TREATING FIELDS

Abstract

A computer-implemented method to determine placement of transducers on a subject's body, the method including: selecting a plurality of intersecting line segment pairs on an image of the subject's body, each of the line segment pairs intersecting in a region of the image corresponding to a tumor in the subject's body, each of the line segment pairs corresponding to locations to place the transducers on the subject's body; determining a pair value for each of the intersecting line segment pairs, each pair value based on a length of each line segment of the corresponding intersecting line segment pair; selecting one or more intersecting line segment pairs based on the pair values to obtain one or more selected intersecting line segment pairs; and outputting the locations to place the transducers on the subject's body corresponding to the one or more selected intersecting line segment pairs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. patent application Ser. No. 17/517,407 filed Nov. 2, 2021, which claims priority to U.S. Patent Application No. 63/110,674 filed Nov. 6, 2020, which are incorporated herein by reference.

BACKGROUND

[0002] Tumor treating fields (TTFields) are low intensity alternating electric fields within the intermediate frequency range, which may be used to treat tumors as described in U.S. Pat. No. 7,565,205. TTFields are induced non-invasively into a region of interest by transducers placed on the patient's body and applying AC voltages between the transducers. Conventionally, a first pair of transducers and a second pair of transducers are placed on the subject's body. AC voltage is applied between the first pair of transducers for a first interval of time to generate an electric field with field lines generally running in the front-back direction. Then, AC voltage is applied at the same frequency between the second pair of transducers for a second interval of time to generate an electric field with field lines generally running in the right-left direction. The system then repeats this two-step sequence throughout the treatment.

SUMMARY OF THE INVENTION

[0003] One aspect of the invention is directed to a computer-implemented method of determining placement of transducers on a subject's body. The computer includes one or more processors and memory accessible by the one or more processors, the memory storing instructions that when executed by the one or more processors cause the computer to perform the method. The method includes selecting a plurality of intersecting line segment pairs on an image of the subject's body, each of the line segment pairs intersecting in a region of the image corresponding to a tumor in the subject's body, each of the line segment pairs corresponding to locations to place the transducers on the subject's body; determining a pair value for each of the intersecting line segment pairs, each pair value based on a length of each line segment of the corresponding intersecting line segment pair; selecting one or more intersecting line segment pairs based on the pair values to obtain one or more selected intersecting line segment pairs; and outputting the locations to place the transducers on the subject's body corresponding to the one or more selected intersecting line segment pairs.

[0004] The above aspect of the invention is exemplary, and other aspects and variations of the invention will be apparent from the following detailed description of embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a flowchart depicting an example of determining placement of transducers on a subject's body.

[0006] FIG. 2 illustrates an example of a subject's body in which an intersecting line segment pair

passes through a tumor.

[0007] FIG. **3** illustrates an example of a subject's body in which intersecting line segment pairs pass through a tumor.

[0008] FIG. **4** illustrates an example magnetic resonance imaging (MRI) image of a subject's head in which an intersecting line segment pair passes through different tissue types and a tumor.

[0009] FIG. **5** illustrates an example MRI image of a subject's torso in which an intersecting line segment pair passes through different tissue types and a tumor.

[0010] FIGS. **6A** and **6B** illustrate example graphs comparing a calculation of the LMIPD to exemplary embodiments of the invention.

[0011] FIGS. **7A-7D** illustrate examples of the structure of various transducers.

[0012] FIG. **8** illustrates an example of a configuration of a pair of transducers.

[0013] FIG. **9** illustrates an example of an apparatus to determine placement of transducers on a subject's body.

DESCRIPTION OF EMBODIMENTS

[0014] To provide a subject with the an effective TTFields treatment, precise locations at which to place the transducers on the subject's body must be generated, and these precise locations are based on the type of cancer and the location of the cancer in the subject's body. However, determining these precise locations is very challenging and involves lengthy and resource intensive computer simulations of numerous possible locations to place the transducers.

[0015] One difficulty in these computer simulations is to account for the conductivities of the different types of tissue (e.g., bone, organs, fluid, skin, and tumor) in the computer simulations. A further difficulty is in modeling higher resolution images for computer simulations, resulting in more complex computer models of the subject's body. As a result, detailed computer simulations of the subject's body for numerous possible locations of the transducers are very costly in terms of computational resources and time.

[0016] The inventors recognized these problems and discovered an approach to determine precise locations at which to place the transducers on the subject's body without costly simulations. In particular, the positions at which to place transducers on a subject's body may be determined based on relationships between the transducers used to induce TTFields. The relationships between the transducers may be based on, for example, the distance between the transducers, the pixels between the transducers in an image of the subject's body, and/or the tissue between the transducers in an image of the subject's body.

[0017] FIG. **1** is a flowchart depicting an example of determining placement of transducers on a subject's body. In one embodiment, each transducer may be an array of electrode elements, and each line segment may thus represent a distance between center points of a pair of transducer arrays.

[0018] At step **110**, the method **100** may select a plurality of intersecting line segment pairs on an image of the subject's body. In one embodiment, each intersecting line segment pair may have a first line segment and a second line segment. The image of the subject's body may include a region associated with a tumor in the subject's body. The image of the subject's body may, for example, be an X-ray image, a computerized tomography (CT) image, a magnetic resonance imaging (MRI) image, or an ultrasound image of the subject's body, or any image of the subject's body providing an internal view of the subject's body. The image may be a so-called slice through the subject's body obtained by scanning equipment.

[0019] Each line segment of each intersecting line segment pair may identify locations to place a pair of transducers on the subject's body. Each pair of transducers may correspond to a channel for generating TTFields in the subject's body. A particular line segment may be used in only one or more than one intersecting line segment pair. Each intersecting line segment pair intersects in a region corresponding to a tumor in the subject's body.

[0020] Each line segment may represent a distance between two transducers and may be defined

by, for example, a point on a first transducer and a point on a second transducer, an intersection with the first transducer and an intersection with the second transducer, pixels of the image, and/or voxels of the image.

[0021] Each line segment of the pair of intersecting line segments may intersect at an intersection angle in the region of the image corresponding to the tumor in the subject's body. For example, the line segment pair may intersect within the tumor in the image, at a centroid of the tumor in the image, or at a point adjacent to the tumor in the image. The line segments of each intersecting line segment pair may be substantially perpendicular, intersect at an angle based on the physical geometry of the subject's body, or intersect at an angle based on the type of transducer to be used on the subject's body. In some cases, an intersection angle within $90^\circ \pm 15^\circ$ may be required.

[0022] At step **120**, the method **100** may determine a pair value for each of the intersecting representative line segment pairs. In one embodiment, the pair value may be based on a length of each line segment of the corresponding intersecting line segment pair. In one example, the pair value of the intersecting representative line segment pair may be calculated, for example, as an absolute value of a difference between the lengths of the line segments in the intersecting line segment pair. In another example, the pair value of the intersecting representative line segment pair may be calculated as a summation of the lengths of the line segments in the intersecting line segment pair.

[0023] In one embodiment, the length of each representative line segment may be calculated based on a distance (e.g. mm) between the endpoints of the representative line segment in the image, a geometric distance (e.g., mm) of each representative line segment, or the relative units of the image. In one example, the pair value T of the intersecting representative line segment pair may be calculated by the following equation:

$$[00001] \ T = \text{.Math. } d1 - d2 \text{ .Math.} \quad \text{Equation1}$$

where d1 and d2 are the distances of the first line segment and the second line segment of the intersecting representative line segment pair.

[0024] In another example, the pair value T for each intersecting representative line segment pair may be calculated by the following equation:

$$[00002] \ T = d1 + d2 \quad \text{Equation2}$$

where d1 and d2 are the distances of the first line segment and the second line segment of the intersecting representative line segment pair.

[0025] In another embodiment, the length of each representative line segment may be calculated based on a number of pixels or a number of voxels in the image between the endpoints of the representative line segment. In a more specific example, the method may identify pixels of the image with which each line segment of each of the intersecting line segment pairs intersects. In another example, the method further comprises assigning pixel tissue values to pixels of the image based on tissue types of the subject's body and determining the pair value based on the pixel tissue values for the pixels intersecting each line segment of the intersecting line segment pair.

[0026] In one example, the pair value T for each intersecting representative line segment pair may be calculated by the following equation:

[00003]

$$T = \text{.Math. } d1 - d2 \text{ .Math.} = \text{.Math. } \text{.Math.}^{N_1} d1_p - \text{.Math.}^{N_2} d2_v \text{ .Math.} \quad \text{Equation3}$$

where the first line segment of the intersecting line segment pair has N.sub.1 pixels d1.sub.p and the second line segment of the intersecting line segment pair has N.sub.2 pixels d2.sub.v.

[0027] In another example, the pair value T for each intersecting representative line segment pair may be calculated by the following equation:

$$[00004] \ T = d1 + d2 = \text{.Math.}^{N_1} d1_p + \text{.Math.}^{N_2} d2_v \quad \text{Equation4}$$

where the first line segment of the intersecting line segment pair has N.sub.1 pixels d1.sub.p and

the second line segment of the intersecting line segment pair has $N_{\text{sub}.2}$ pixels $d_{2,\text{sub}.v}$.

[0028] In another embodiment, each pair value may be based on the weighted distance between the first endpoint and the second endpoint of each line segment of the corresponding intersecting line segment pair. In one example, the weighted distance between the first endpoint and the second endpoint of each line segment is based on one or more tissue types within the portion of the subject's body through which the corresponding line segment passes. In a more specific example, the method further comprises assigning pixel tissue weightings to pixels of the image of the subject's body based on tissue types of the subject's body. In one example, the tissue types of the subject's body comprise gray matter, white matter, and bone. In another example, the tissue types of the subject's body comprise organ tissue, muscular tissue, and bone. The weight may be based on, for example, the tissue type conductivity or resistivity.

[0029] In one example, the pair value T for each intersecting representative line segment pair may be calculated by the following equation:

[00005]

$$T = |d_1 - d_2| = \sum_{p=1}^{N_1} w_{1,p} d_{1,p} - \sum_{v=1}^{N_2} w_{2,v} d_{2,v} \quad \text{Equation 5}$$

where the first line segment of the intersecting line segment pair has $N_{\text{sub}.1}$ pixels $d_{1,\text{sub}.p}$, the second line segment of the intersecting line segment pair has $N_{\text{sub}.2}$ pixels $d_{2,\text{sub}.v}$, the pixel $d_{1,\text{sub}.p}$ has a tissue weight $w_{1,\text{sub}.p}$, and the pixel $d_{2,\text{sub}.v}$ has a tissue weight $w_{2,\text{sub}.v}$.

[0030] In another example, the pair value T for each intersecting representative line segment pair may be calculated by the following equation:

$$[00006] \quad T = d_1 + d_2 = \sum_{p=1}^{N_1} w_{1,p} d_{1,p} + \sum_{v=1}^{N_2} w_{2,v} d_{2,v} \quad \text{Equation 6}$$

where the first line segment of the intersecting line segment pair has $N_{\text{sub}.1}$ pixels $d_{1,\text{sub}.p}$, the second line segment of the intersecting line segment pair has $N_{\text{sub}.2}$ pixels $d_{2,\text{sub}.v}$, the pixel $d_{1,\text{sub}.p}$ has a tissue weight $w_{1,\text{sub}.p}$, and the pixel $d_{2,\text{sub}.v}$ has a tissue weight $w_{2,\text{sub}.v}$.

[0031] At step **130**, the method **100** may select one or more intersecting representative line segment pairs based on the pair values determined at step **120**. For example, if an absolute value calculation is used to determine the pair values, the pair values may be sorted, the smaller pair values or the smallest pair value may be determined, and the intersecting representative line segment pairs corresponding to the smaller pair values or the smallest pair value may be selected. As another example, if a summation calculation is used to determine the pair values, the pair values may be sorted, the smaller pair values or the smallest pair value may be determined, and the intersecting representative line segment pairs corresponding to the smaller pair values or the smallest pair value may be selected.

[0032] As another example, the intersecting representative line segment pairs may be selected based on comparing a threshold to the pair values of the intersecting line segment pairs. For example, if an absolute value calculation is used to determine the pair values, the intersecting line representative segment pairs corresponding to pair values at or below the threshold may be selected. As another example, if a summation calculation is used to determine the pair values, the intersecting representative line segment pairs corresponding to pair values at or below the threshold may be selected.

[0033] In one embodiment, at least one of the one or more selected intersecting representative line segment pairs may have a local minimum power density (LMiPD) at an intersection of the line segments that is higher than an LMiPD at an intersection of the line segments of non-selected intersecting line segment pairs. The LMiPD may represent the minimum dose delivered by the TTFields to the tumor via a particular transducer layout, and an ideal transducer layout may be obtained when this minimum dose is maximized relative to other potential layouts.

[0034] In another embodiment, the one or more selected intersecting line segment pairs are selected without simulating TTFields for transducer locations on the subject's body.

[0035] At step **140**, the method **100** may output the locations to place the transducers on the

subject's body corresponding to the one or more selected intersecting representative line segment pairs. The output may be sent to a user device. In one embodiment, the locations to place the transducers on the subject's body are outputted without simulating the TTFields for the locations. [0036] FIG. 2 illustrates an example portion of a subject's body in which an intersecting line segment pair passes through a tumor. In the example depicted in FIG. 2, the image **200** of the subject's body includes a tumor **205**, and a first line segment **250a** and a second line segment **250b** intersecting in the tumor **205**. The first line segment **250a** corresponds to the locations of a first transducer **210** and a second transducer **220**. In one example, the endpoints of the first line segment **250a** may correspond to the locations of the centers of the transducers **210** and **220**. The second line segment **250b** corresponds to the locations of a third transducer **230** and a fourth transducer **240**. In one example, the endpoints of the second line segment **250b** may correspond to the locations of the centers of the transducers **230** and **240**. The first line segment **250a** and the second line segment **250b** form an intersecting line pair **250**. The first line segment **250a** and the second line segment **250b** intersect at substantially 90°.

[0037] The line segment **250a** may have a length between the first transducer **210** and the second transducer **220** represented by distance (or length) d_1 , and line segment **250b** may have a length between the third transducer **230** and the fourth transducer **240** represented by distance (or length) d_2 . In one embodiment, the pair value of the intersecting line pair **250** may be calculated based on Equations 1-4 discussed above.

[0038] FIG. 3 illustrates an example portion of a subject's body in which intersecting line segment pairs pass through a tumor. In the example depicted in FIG. 3, the image **200** of the subject's body includes three intersecting line segment pairs representing the locations of three pairs of transducers. For clarity, the three pairs of transducers are not shown in FIG. 3. In particular, the image **200** of the subject's body includes the first intersecting line segment pair **250** of the line segments **250a** and **250b**, a second intersecting line segment pair **320** of line segments **320a** and **320b**, and a third intersecting line segment pair **330** of line segments **330a** and **330b**. Each of the intersecting line segment pairs intersects within the tumor **205**.

[0039] Potential transducer locations on the subject's body may be spaced apart by predetermined angles. As an example, the first line segments of each intersecting representative line segment pair may be spaced apart by a predetermined angle, and the second line segments of each intersecting line segment pair may be spaced apart by the predetermined angle. The predetermined angle may be, for example, 0.5°, 1°, 5°, 10°, 15°, 30°, 45°, 60°, 90°, or any other angle. As another example, the line segments of each intersecting line segment pair may be spaced apart by different angles. In the example depicted in FIG. 3, the first line segment **250a** of the first intersecting line segment pair **250** may be spaced apart by a predetermined angle from the first line segment **320a** of the second intersecting line segment pair **320**, and the first line segment **320a** of the second intersecting line segment pair **320** may be spaced apart by the predetermined angle from the first line segment **330a** of the third intersecting line segment pair **330**. Similarly, the second line segment **250b** of the first intersecting line segment pair **250** may be spaced apart by the predetermined angle from the second line segment **320b** of the second intersecting line segment pair **320**, and the second line segment **320b** of the second intersecting line segment pair **320** may be spaced apart by the predetermined angle from the second line segment **330b** of the third intersecting line segment pair **330**.

[0040] FIG. 4 illustrates an example magnetic resonance imaging (MRI) image of a subject's head in which an intersecting line segment pair passes through different tissue types and a tumor. In the example depicted in FIG. 4, MRI image **400** of a subject's head includes a tumor **405**, an intersecting representative line segment pair **450** passes through different tissue types and intersects with the tumor **405**. The intersecting representative line segment pair **450** includes a first line segment **450a** and a second line segment **450b**. The first line segment **450a** is defined by the locations of the transducers **410** and **420**, and the second line segment **450b** is defined by the

locations of the transducers **430** and **440**. The tissue types of the subject's body may include gray matter **401**, white matter **402**, bone **403**, brain fluid, and skin. In FIG. **4**, the first line segment **450a** and the second line segment pass through multiple tissue types (e.g., skin, bone, brain fluid, white matter, and/or gray matter). The distance d_1 of line segment **450a** may include multiple pixels passing through skin, bone, brain fluid, white matter, and gray matter, with each pixel being weighted accordingly. Then, d_1 may be the summation of each weighted pixel along the line segment **450a**. Similarly, the distance d_2 of line segment **450b** may be the summation of each weighted pixel along the line segment **450b**. The pair value of the intersecting line pair **450** may be calculated based on Equations 5-6 discussed above.

[0041] FIG. **5** illustrates an example MRI image of a subject's torso in which an intersecting line segment pair passes through different tissue types and a tumor. In the example depicted in FIG. **5**, the MRI image **500** of the subject's torso includes a tumor **505** and an intersecting representative line segment pair **550** passing through different tissue types and intersecting with the tumor **505**. The tissue types of the subject's body may include organ tissue **501**, muscular tissue **502**, bone **503**, skin, and fluid. The intersecting representative line segment pair **550** includes a first line segment **550a** and a second line segment **550b**. The first line segment **550a** is defined by the locations of the transducers **510** and **520**, and the second line segment **550b** is defined by the locations of the transducers **530** and **540**.

[0042] FIGS. **6A** and **6B** illustrate example graphs comparing a calculation of the local minimum power density (LMiPD) to exemplary embodiments of the invention.

[0043] In FIGS. **6A** and **6B**, the x-axis corresponds to angles of a channel (CH0) for a pair of transducers on a subject's head, where 0° refers to the front-back line of the subject's head and where the angles are with respect to this front-back line, the left y-axis corresponds to power loss units (mW/cm.sup.3), and the right y-axis corresponds to normalized weighted distance between the pair of transducers. In one example, a weight w with a relative value between -3 and 3 may be assigned to a pixel based on its respective tissue type. The graph includes the power loss for the channel (CH0) for the pair of transducers positioned in a front-to-back orientation on the subject's head and then shifted by the angle of the x-axis. The angles for the data points of the channel (CH0) are between 0° and 170° in 5° increments.

[0044] In the example depicted in FIG. **6A**, graph **601A** corresponds to the power loss at the target region (e.g., tumor) calculated based on the weighted distance between the pair of transducers on the subject's body. Graph **602A** corresponds to the power loss at a tumor in the subject's head calculated based on a complex simulation calculating LMiPD. As shown in FIG. **6A**, the transducer layouts identified by calculating the weighted distance using exemplary embodiments of the invention are corroborated by the complex simulations to calculate the LMiPD.

[0045] In the example depicted in FIG. **6B**, graph **601B** corresponds to the power loss at the target region (e.g., tumor) calculated based on weighted distance between a pair of transducers on the subject's body. Graph **602B** corresponds to the power loss at a tumor in the subject's head calculated based on a complex simulation calculating LMiPD. As shown in FIG. **6B**, the transducer layouts identified by calculating the weighted distance using exemplary embodiments of the invention are corroborated by the complex simulations to calculate the LMiPD.

[0046] FIGS. **7A-7D** illustrate examples of the structure of various transducers. In one embodiment, the transducers comprise arrays of substantially flat electrode elements.

[0047] In FIG. **7A**, the transducer **700A** may have a substrate **701A** and a plurality of electrode elements **702A**. The substrate **701A** may be configured for attaching the transducer **700A** to a subject's body. Suitable materials for the substrate **701A** may include, as examples, cloth, foam, and flexible plastic. In one example, the substrate **701A** may include a conductive medical gel. In a more specific example, the substrate **701A** may be a layer of hydrogel.

[0048] A plurality of capacitively coupled electrode elements **702A** may be positioned on the substrate **701A**, and each of the capacitively coupled electrode elements may have a conductive

plate with a dielectric layer disposed thereon that faces towards the substrate. Optionally, one or more sensors may be positioned beneath each of the electrode elements in a manner that is similar to the conventional arrangement used in the Novocure Optune® system. In one example, the one or more sensors may be temperature sensors (e.g., thermistors).

[0049] FIG. 7B depicts another example of the structure of the transducer **700B**. In this example, the transducer **700B** may include a plurality of electrode elements **702B**. The plurality of electrode elements **702B** may be electrically and mechanically connected to one another without a substrate. In one example, the electrode elements **702B** may be connected to one another through conductive wires **701B**.

[0050] FIGS. 7C and 7D are further examples of the structure of the transducers **700C** and **700D**. For example, FIG. 7C depicts an example of a transducer **700C** having an array of thirteen electrode elements **702C** disposed on the substrate **701C**. Furthermore, FIG. 7D depicts an example of a transducer **700D** having an array of twenty electrode elements **702D** disposed on the substrate **701D**.

[0051] In one example, the electrode elements **702A**, **702B**, **702C**, and **702D** may be ceramic disks, and each of the ceramic disks may be approximately 2 cm in diameter and approximately 1 mm in thickness. In another example, the electrode elements **702A**, **702B**, **702C**, and **702D** may be ceramic elements that are not disk-shaped. In yet another example, the electrode elements **702A**, **702B**, **702C**, and **702D** may be non-ceramic dielectric materials positioned over a plurality of flat conductors. Examples of non-ceramic dielectric materials positioned over flat conductors may include polymer films disposed over pads on a printed circuit board or over flat pieces of metal. In particular embodiments, transducers that use an array of electrode elements that are not capacitively coupled may also be used. In this situation, each electrode element **702A**, **702B**, **702C**, **702D** may be implemented using a region of a conductive material that is configured for placement against a subject's body, with no insulating dielectric layer disposed between the conductive elements and the body. In other embodiments, the transducer may include only a single electrode element. As an example, the single electrode element may be a flexible organic material or flexible organic composite positioned on a substrate. As another example, the transducer may include a flexible organic material or flexible organic composite without a substrate.

[0052] Other alternative constructions for implementing a transducer for use with embodiments of the invention may also be used, as long as they are capable of (a) delivering TTFields to the subject's body and (b) being positioned at the locations specified herein.

[0053] FIG. 8 illustrates an example of a configuration of a pair of transducers. In this example, the first transducer **801** may include thirteen electrode elements **803** which are positioned on the substrate **804**, and the electrode elements **803** may be electrically and mechanically connected to one another through a conductive wiring **809**. Similarly, the second transducer **802** may include twenty electrode elements **805** which are positioned on the substrate **806**, and the electrode elements **805** may similarly be electrically and mechanically connected to one another through a conductive wiring **810**. Furthermore, the first transducer **801** and the second transducer **802** may be connected to an AC voltage generator **807** and a controller **808**. The controller **808** may include one or more processors and memory accessible by the one or more processors. The memory may store instructions that, when executed by the one or more processors, control the AC voltage generator **807** to implement a first electric field between the first pair of transducers **801**, **802**, then implement a second electric field between a second pair of transducers (not shown), and then alternately iterate between implementing the first electric field and the second electric field. As shown in FIG. 8, the transducers **801**, **802** are different. The transducers **801**, **802** may be the same or may differ in terms of, for example, number of elements and/or locations of the elements.

[0054] FIG. 9 illustrates an example of an apparatus to determine locations of transducers on a subject's body using the exemplary embodiments discussed herein. In this example, the apparatus **900** may include one or more processors **902**, one or more output devices **905**, and a memory **903**.

[0055] In one embodiment, the one or more processors **902** may include a general purpose processor, an integrated circuit, a server, other programmable logic device, or any combination thereof. The processor may be a conventional processor, microprocessor, controller, microcontroller, or state machine. The one or more processors may be one, two, or more processors of the same or different types. Furthermore, the one or more processors may be a computer, computing device and user device, and the like.

[0056] In one example, based on user input **901**, the one or more processors may determine positions at which to place transducers on a subject's body based on relationships between channels used to induce TTFields and may make one or more recommendations to the user. The one or more recommendations may be output on the one or more output devices **905**. In another example, the user may give feedback regarding the one or more recommendations through the output devices **905**. After receiving the feedback from the user, the one or more processors **902** may generate one or more different recommendations regarding the locations of the transducers.

[0057] The memory **903** may be accessible by the one or more processors **902** via the link **904** so that the one or more processors **902** can read information from and write information to the memory **903**. Memory **903** may be integral with or separate from the processors. Examples of the memory **903** include RAM, flash, ROM, EPROM, EEPROM, registers, disk storage, or any other form of storage medium. The memory **903** may store instructions that, when executed by the one or more processors **902**, implement one or more embodiments of the invention. Memory **903** may be a non-transitory computer-readable medium that stores instructions, which when executed by a computer, cause the computer to perform one or more of the exemplary methods discussed herein.

[0058] Numerous modifications, alterations, and changes to the described embodiments are possible without departing from the scope of the present invention defined in the claims. It is intended that the present invention not be limited to the described embodiments, but that it has the full scope defined by the language of the following claims, and equivalents thereof.

Claims

1. A method, machine, manufacture, and/or system substantially as shown and described.
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